

K.R. MANGALAM UNIVERSITY
SCHOOL OF ENGINEERING AND TECHNOLOGY
B.TECH CSE (AI & ML)

IMPLEMENTATION OF XOR, MUX AND DMUX CHIPS USING HDL
NAND2TETRIS HARDWARE SIMULATOR

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Objective:

To implement XOR, 2-way MUX, 4-way 16-bit MUX, and DMux chips using Nand2Tetris HDL, construct their truth tables, and verify the implementations using the corresponding Nand2Tetris test scripts.

1. XOR Gate

The XOR gate gives output 1 when exactly one of the two inputs is 1.

A	B	OUT
0	0	0
0	1	1
1	0	1
1	1	0

HDL implementation:

```
CHIP Xor {
    IN a, b;
    OUT out;

    PARTS:
    Or(a=a, b=b, out=orAB);
    And(a=a, b=b, out=andAB);
    Not(in=andAB, out=notAnd);
    And(a=orAB, b=notAnd, out=out);
}
```

Test file: Xor.tst. The test script loads Xor.hdl, compares the output with Xor.cmp, and evaluates the input combinations.

The screenshot displays the NAND2Tetris Hardware Simulator interface. The main window shows the HDL code for a chip named `DMux`. The code implements a demultiplexer that takes an input `in` and a selector `sel`, and produces two outputs `a` and `b`. The implementation uses internal components: a `Not` gate, an `And` gate, and a `DMux` chip (which is a recursive call to itself). The test script at the bottom shows the simulation setup, including loading the HDL file, comparing it to a reference file, and evaluating the results. The status bar at the bottom indicates that the simulation was successful.

```
4 // File name: projects/1/DMux.hdl
5 /**
6  * Demultiplexor:
7  * [a, b] = [in, 0] if sel = 0
8  *           [0, in] if sel = 1
9  */
10 CHIP DMux {
11     IN in, sel;
12     OUT a, b;
13
14     PARTS:
15     Not(in=sel, out=notSel);
16     And(a=in, b=notSel, out=a);
17     And(a=in, b=sel, out=b);
18 }
```

Test Script

```
6 load DMux.hdl,
7 compare-to DMux.cmp,
8 output-list in sel a b;
9
10 set in 0,
11 set sel 0,
12 eval,
```

Simulation successful: The output file is identical to the compare file

Figure 1: XOR HDL implementation and test script evidence.

2. 2-Way Multiplexer (MUX)

A 2-way MUX selects one of two inputs. When sel = 0, OUT = A; when sel = 1, OUT = B.

A	B	SEL	OUT
0	0	0	0
0	1	0	0
1	0	0	1
1	1	0	1
0	0	1	0
0	1	1	1
1	0	1	0
1	1	1	1

HDL implementation:

```
CHIP Mux {
  IN a, b, sel;
  OUT out;

  PARTS:
    Not(in=sel, out=notSel);
    And(a=a, b=notSel, out=aPart);
    And(a=b, b=sel, out=bPart);
    Or(a=aPart, b=bPart, out=out);
}
```

Test file: Mux.tst. The test script loads Mux.hdl, compares the output with Mux.cmp, and evaluates the required combinations.

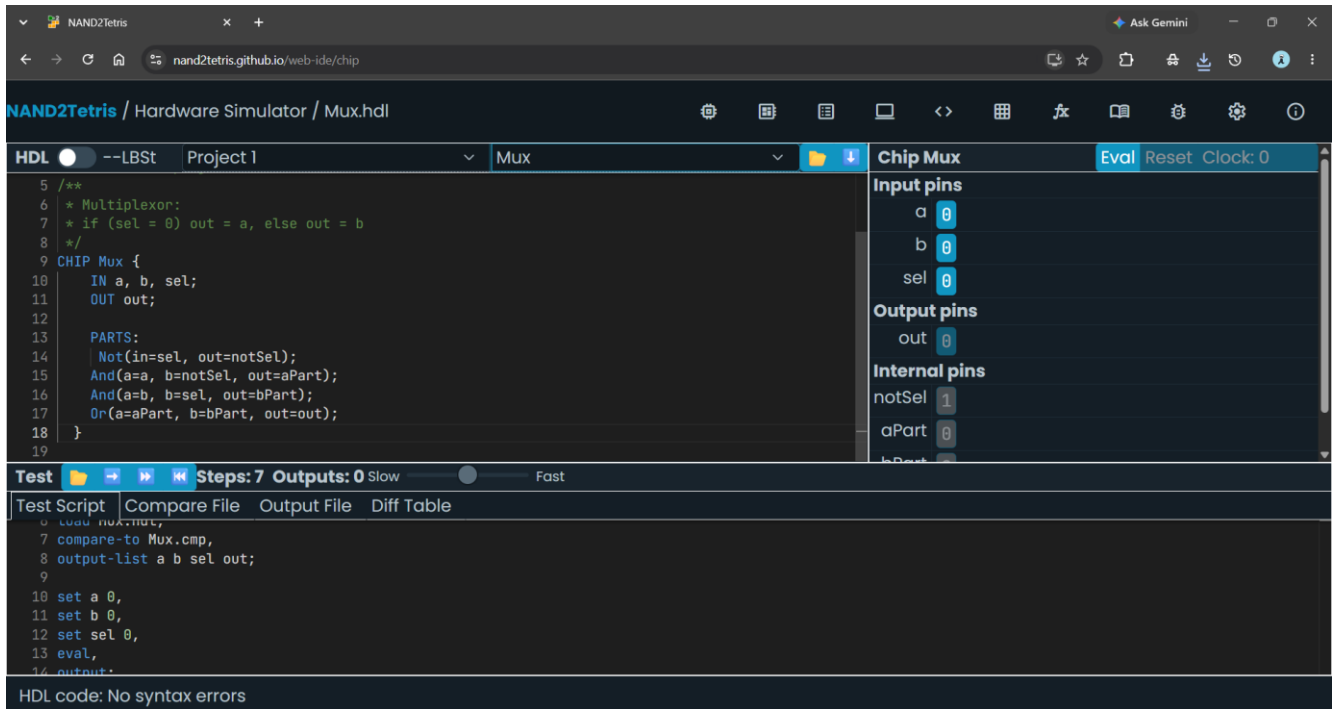


Figure 2: 2-way MUX HDL implementation and test script evidence.

3. 4-Way 16-Bit Multiplexer (Mux4Way16)

Mux4Way16 selects one of four 16-bit inputs using two select bits.

SEL[1]	SEL[0]	Selected Input
0	0	A
0	1	B
1	0	C
1	1	D

HDL implementation:

```

CHIP Mux4Way16 {
    IN a[16], b[16], c[16], d[16], sel[2];
    OUT out[16];

    PARTS:
        Mux16(a=a, b=b, sel=sel[0], out=ab);
        Mux16(a=c, b=d, sel=sel[0], out=cd);
        Mux16(a=ab, b=cd, sel=sel[1], out=out);
}

```

Test file: Mux4Way16.tst. The test script loads Mux4Way16.hdl, compares the output with Mux4Way16.cmp, and tests all four select combinations.

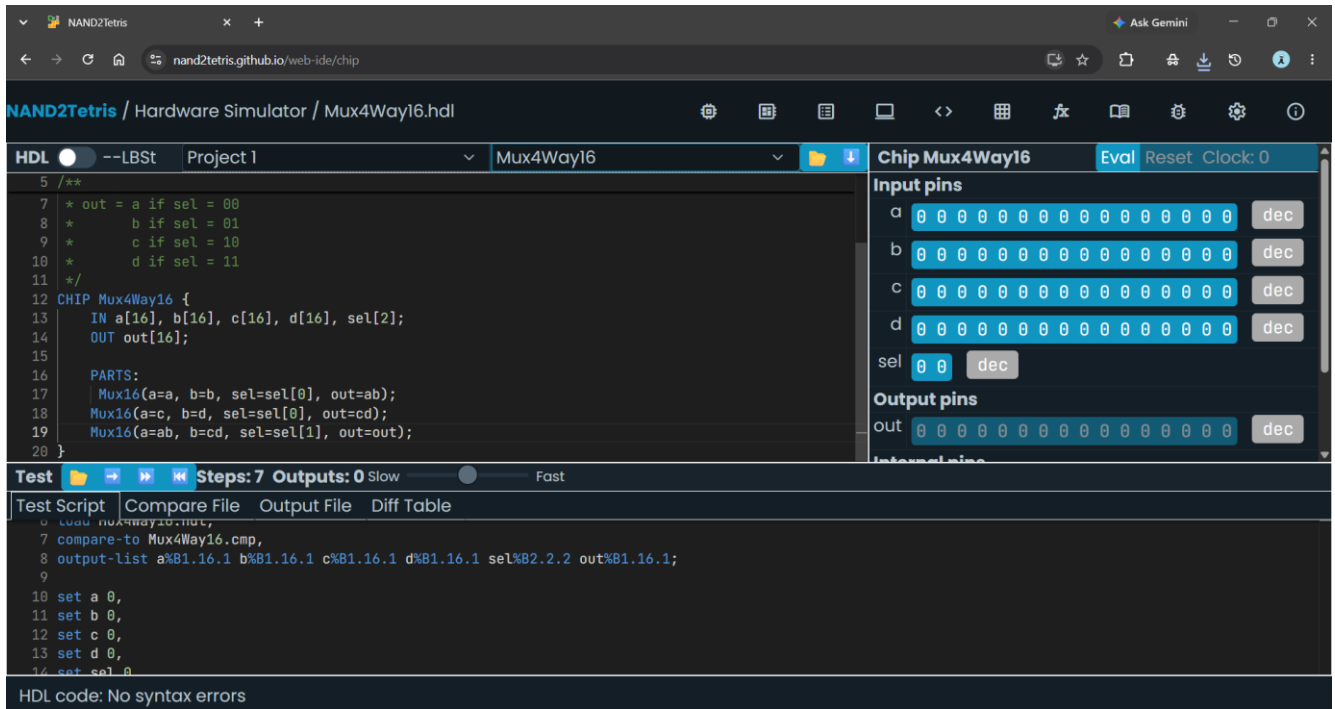


Figure 3: 4-way 16-bit MUX HDL implementation and test script evidence.

4. Demultiplexer (DMux)

The DMux routes one input to one of two outputs. When $sel = 0$, the input is routed to A; when $sel = 1$, it is routed to B.

IN	SEL	A	B
0	0	0	0
0	1	0	0
1	0	1	0
1	1	0	1

HDL implementation:

```

CHIP DMux {
    IN in, sel;
    OUT a, b;

    PARTS:
        Not(in=sel, out=notSel);
        And(a=in, b=notSel, out=a);
        And(a=in, b=sel, out=b);
}

```

Test file: DMux.tst. The test script loads DMux.hdl, compares the output with DMux.cmp, and verifies the expected results.

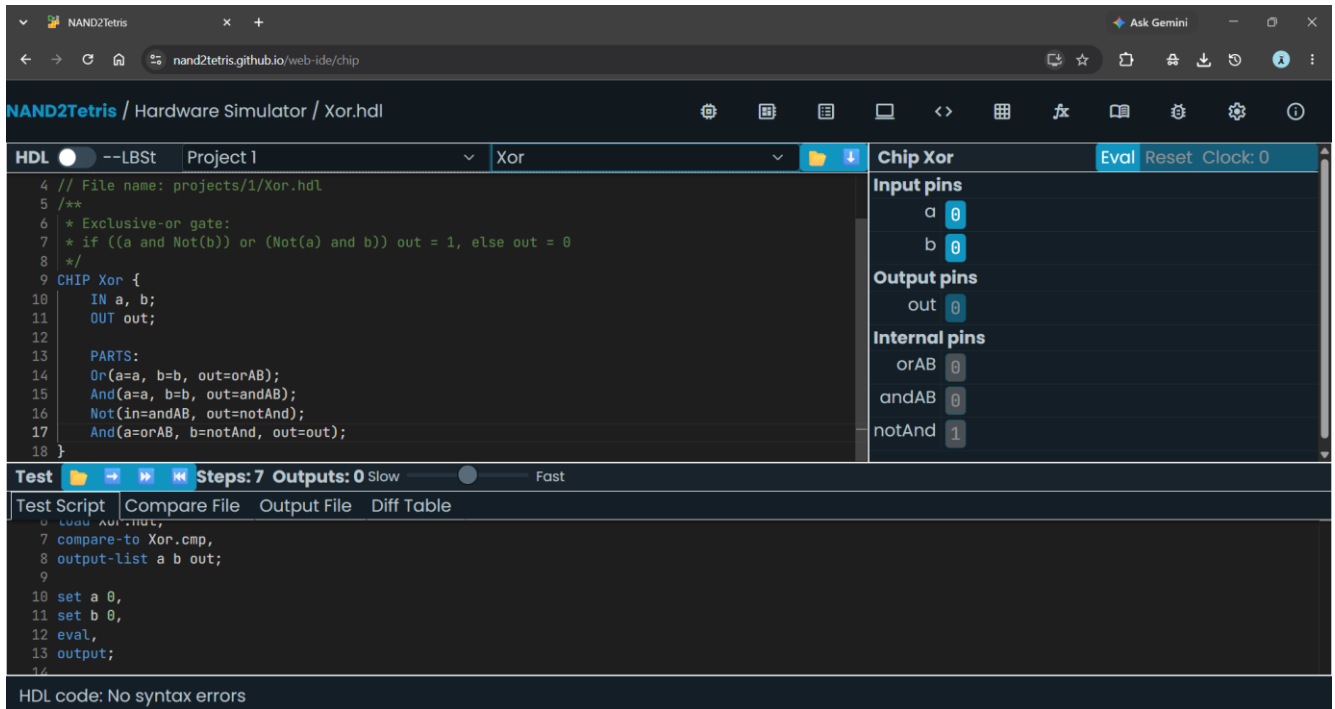


Figure 4: DMux HDL implementation and successful simulation evidence.

5. Verification Summary

Chip	HDL File	Test File	Evidence
XOR	Xor.hdl	Xor.tst	Simulator screenshot
2-way MUX	Mux.hdl	Mux.tst	Simulator screenshot
4-way 16-bit MUX	Mux4Way16.hdl	Mux4Way16.tst	Simulator screenshot
DMux	DMux.hdl	DMux.tst	Simulation successful

6. Conclusion

The XOR, 2-way MUX, 4-way 16-bit MUX, and DMux chips were implemented using Nand2Tetris HDL. Their truth tables, HDL implementations, test scripts, and simulator screenshots are included as evidence of implementation and verification.